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THEORETICAL TOOLS IN DB RESEARCH

Theoretical Tools in DB research

Lecture notes distilled from the SIGMOD 2026 full-corpus survey

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Introduction: What DB Researchers Prove, and How

0.0 What these notes are

These lecture notes are a *toolbox tour* of the theoretical machinery that database researchers actually use, extracted mechanically from the complete proceedings of **SIGMOD 2026**: 385 research papers (PACMMOD Vol. 4, rounds 3–4, plus Vol. 3 issues 4–6, rounds 1–2). Every paper was fetched (arXiv $\text{\LaTeX}$ sources when available, ACM camera-ready PDFs otherwise), machine-extracted into formal statements, proofs, and bound sentences (2,405 statements, 1,130 proofs), and 228 theory-relevant papers were tagged with a controlled vocabulary of properties, tools, domains, and roles. The aggregates behind every figure in this chapter come from that survey; every worked example in the later chapters is a real paper from the corpus, cited in that chapter’s reference list.

Two honesty markers used throughout:

- $\text{\texttt{PODS}}$ marks exemplars from the **PODS’26 issue** (PACMMOD Vol. 4 No. 2, DOI band 10.1145/3801890–3801919): 29 of the 228 tagged papers (12.7%). They are part of the proceedings bundle; they often provide particularly clean exhibits of a tool, but they carry the theory-first culture of a sibling conference.
- Formal statements are transcribed from machine extraction of the papers; where the extraction was noisy we paraphrase conservatively rather than decorate.

0.1 The survey questions

1. **Modeling** — how do papers formalize a systems problem (cost model, adversary, noise model, distance metric)?

2. **Properties** — what do they prove about the artifact: correctness, complexity bounds, approximation/competitive ratios, statistical guarantees?
3. **Tools** — which theoretical machinery delivers the proof?
4. **Role** — does theory motivate the work, drive the design, serve as the evaluation yardstick, or *is* it the contribution?

0.2 *Properties: first correct, then fast*

Figure 1 (left) ranks what the 228 tagged papers prove. The headline surprise of the full-corpus count: **correctness is #1** (98 papers, 43%) — ahead of upper bounds (75) and probabilistic guarantees (42). The typical SIGMOD theory section is not a big theorem; it is a small invariant certifying that a component preserves a property (monotonicity of timestamps, order-independence of index construction, merge-safety of compressed blocks). Hardness (36) plays a different role: the standard *license for heuristics* in graph search, repair, and mining papers. The tail is where culture lives: **tightness** (11 — matching upper and lower bounds, a $5.5\times$ jump over the arXiv-only view of the corpus) and freshly activated slots such as competitive ratio and regret bounds (2 papers each, entering through systems doors: when to adapt, when to stop).

0.3 *Tools: a trinity plus a riser*

Figure 1 (right) ranks the machinery. A *trinity* dominates: **reductions** (52 papers), **concentration inequalities** (39), and **induction** (36) — respectively the tools of the hardness culture, the statistical-certification culture, and the component-correctness culture. The big riser once the ACM-only mass arrived is **exchange-greedy** (4→17 papers): prove a selection problem is a submodular/metric instance, inherit the approximation factor. Chapter 0.7 (the index table) maps each tool to its chapter.

0.4 *Domains and dialects*

Figure 2 (left) shows where theory lives. Graph systems (56 papers) are now the dominant dialect — hardness → branch-and-bound and parameterized algorithms → index-correctness lemmas. Privacy (26) is a near-monoculture of composition accounting (69% of its papers), increasingly in Rényi/f-DP currency. Streaming and sketching splits into a pure-theory wing (matching lower bounds) and estimator systems;

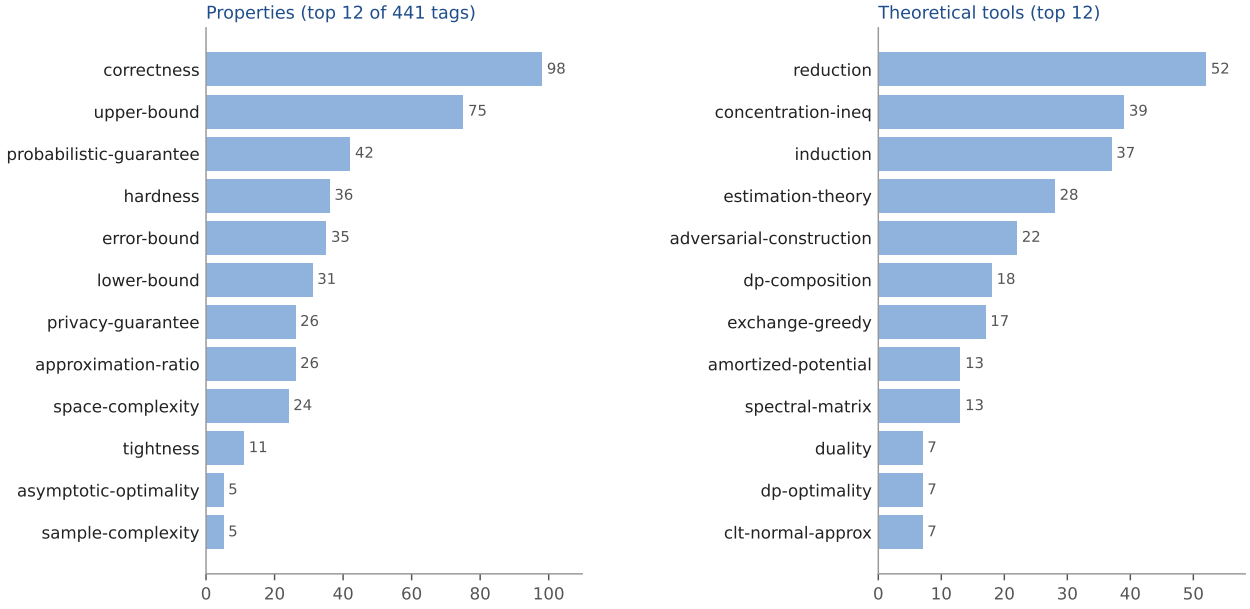


Figure 1: Left: most-studied properties (top 12 of 441 property tags on the 228 tagged papers). Right: most-used theoretical tools (top 12) — each later chapter is one tool.

vector search (25) splits between CLT-style certification and structural correctness of graphs and routers.

0.5 The playbook: *property* × *tool* pairings

Figure 2 (right) is the survey’s central cross-tab: which tool delivers which property. The two workhorses are **hardness** × **reduction** (32 papers) and **correctness** × **induction** (29). The pattern to internalize: DB theory compresses into about eight such pairings, and each chapter of these notes ends with the pairing(s) its tool owns.

0.6 Where theory sits

66% of tagged papers use theory as a *design driver*; in 25% the formal result *is* the paper (concentrated in the PODS issue and the pure streaming enclave); motivation-only and evaluation-yardstick roles cover the rest. The role split matters when reading exemplars: a design-driver paper proves what it builds; a theory-as-contribution paper builds nothing and proves everything.

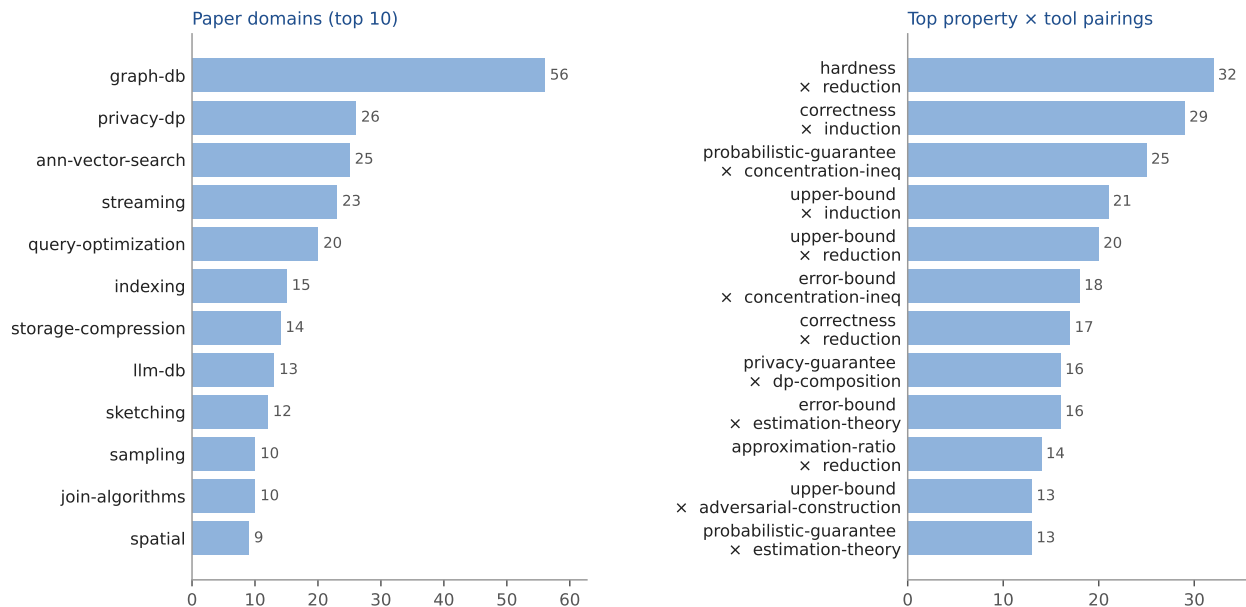


Figure 2: Left: paper domains (top 10). Right: top property $\times$ tool pairings (top 10) — the “playbook”.

0.7 Chapter map

Each of the following chapters is one tool: the formal core, then real SIGMOD 2026 exemplars with minimal context, then a “when to reach for it” box. Exemplars are transcribed from the papers’ own statements, with PODS-issue exemplars capped at one per chapter (two for communication complexity, whose corpus presence is small); a paper may recur in a second chapter when it is a clean exhibit of two tools — each chapter treats it under its own lens, and the ^{PODS} marker identifies PODS-issue exemplars in the text.

Tools not large enough for their own chapter in this corpus but worth knowing by name: adversarial-adjacent duality/LP arguments (7 papers), dynamic programming optimality (7), CLT/normal approximations (7), dyadic peeling (6), sketch mergeability theory (6), balls-bins-hashing (5) — several appear as guest stars inside the thirteen chapters.

0.8 How these notes were made

The survey was scripted end to end: the 385-paper list came from OpenAlex, sources were fetched (arXiv $\text{\LaTeX}$ when available, ACM camera-ready PDFs otherwise), statements, proofs, and bound sentences were machine-extracted (2,405 statements, 1,130 proofs), and the 228 theory-relevant papers were tagged against the controlled vocabulary above;

Table 1: Chapter map. “Papers” = tagged corpus papers using the tool; “Ex.” = exemplars featured in the chapter.

Ch.	Tool	Papers	Ex.
1	Concentration inequalities	38	5
2	Reductions	50	6
3	Induction	36	6
4	Estimation theory	27	6
5	Information-theoretic bounds	5	5
6	Adversarial constructions	21	6
7	Communication complexity	4	4
8	DP composition	18	6
9	Exchange-greedy arguments	16	6
10	Amortized-potential analysis	11	6
11	Spectral / matrix tools	13	6
12	Coresets & randomized NLA	8	6
13	Online decisions (competitive & regret)	4	4

every figure and count in this chapter is a direct aggregate of those tags.

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1

Concentration Inequalities

Concentration inequalities bound the probability that a random quantity strays far from its expectation. For a systems paper they upgrade a randomized component that is correct *in expectation* — a sampler, sketch, random walk, or noise mechanism — into one correct *with high probability (whp)*: a quantified contract such as “ $(1 \pm \varepsilon)$ -accurate with probability $1 - 1/\text{poly}(n)$ ” or “running time $O(T)$ with probability $1 - 1/N$.” The recipe is constant: name the bad event, bound it by a tail inequality, union-bound over all copies of the component. In the SIGMOD’26 corpus this is among the most-used theoretical tools (38 tagged papers), directly supplying the probabilistic-guarantee property in 25 of them.

This chapter presents the basic ladder and the standard exemplars first; Section 1.4 adds the advanced rungs (martingales, two harder exemplars) as extension reading.

1.0 The tool: concentration inequalities

The problem class is uniform across applications: we can compute (or bound) the expectation of some random quantity X — the number of sampled points falling in a region, the error of a private estimate, the load of a hash bucket — but an algorithm must succeed on the *actual* draw, not the average. What concentration delivers is a bound of the form

$$\Pr[|X - \mathbb{E}X| \geq t] \leq \text{some decreasing function of } t,$$

plus the craft of making that function decrease fast enough. How fast is “fast enough” is the organizing question of this chapter: a bound of δ per bad event survives a union bound over $1/\delta$ events, so a component replicated $\text{poly}(n)$ times needs per-event failure $1/\text{poly}(n)$, which in turn needs tails that decay *exponentially* in the deviation.

LEARNING OBJECTIVES

- Recognize when a component needs a *whp* contract, and formalize the bad event.
- Prove the ladder Markov $\rightarrow$ Chebyshev $\rightarrow$ Chernoff/Hoeffding, and match each rung to the dependence structure.
- Tune per-event failure to $\delta/(\text{events})$; union-bound over all copies.
- Dissect a paper’s argument: random object, independence, rung, δ budget.

Notation

“whp” (with high probability) always means probability at least $1 - 1/\text{poly}(n)$, or $1 - 1/N$ for a system of N components; the polynomial in the failure probability is set by the union bound over the whole system.

1.0.1 The basic ladder

Every bound in the family is Markov’s inequality in disguise [2, 4, 13].

Theorem 1.1 — Markov. Let $X \geq 0$ be a random variable with $\mathbb{E}X < \infty$. Then for every $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbb{E}X}{a}. \quad \blacksquare$$

Proof. $\mathbb{E}X \geq \mathbb{E}[X \cdot \mathbf{1}\{X \geq a\}] \geq a \Pr[X \geq a]$; divide by a . $\blacksquare$

One moment, one side, polynomial tail. Applying it to $(X - \mu)^2$ instead of X gives the two-sided, second-moment rung:

Corollary 1.2 — Chebyshev. Let $\mu = \mathbb{E}X$ and $\text{Var}(X) < \infty$. Then for every $t > 0$,

$$\Pr[|X - \mu| \geq t] \leq \frac{\text{Var}(X)}{t^2}. \quad \blacksquare$$

Proof. Markov (Theorem 1.1) applied to the nonnegative variable $(X - \mu)^2$ with $a = t^2$. $\blacksquare$

Polynomial tails suffice for constant success probability — boosted to high probability by repetition and medians — but they cannot survive a union bound over $\text{poly}(n)$ events. Exponential tails come from applying Markov to $e^{\lambda X}$:

Lemma 1.3 — Chernoff method. For any random variable X , any $a \in \mathbb{R}$ and any $\lambda > 0$,

$$\Pr[X \geq a] = \Pr[e^{\lambda X} \geq e^{\lambda a}] \leq e^{-\lambda a} \mathbb{E}[e^{\lambda X}],$$

and one optimizes over $\lambda > 0$ (the *moment generating function* controls the tail). $\blacksquare$

Proof. Markov (Theorem 1.1) applied to the nonnegative variable $e^{\lambda X}$ at level $a' = e^{\lambda a}$; the inequality is monotone in λ ’s sign, so only $\lambda > 0$ is useful. $\blacksquare$

Evaluating the moment generating function of a sum of independent bounded variables yields the two workhorses of randomized systems. The multiplicative form is the one to memorize for sampling and sketching [2, 3, 13]:

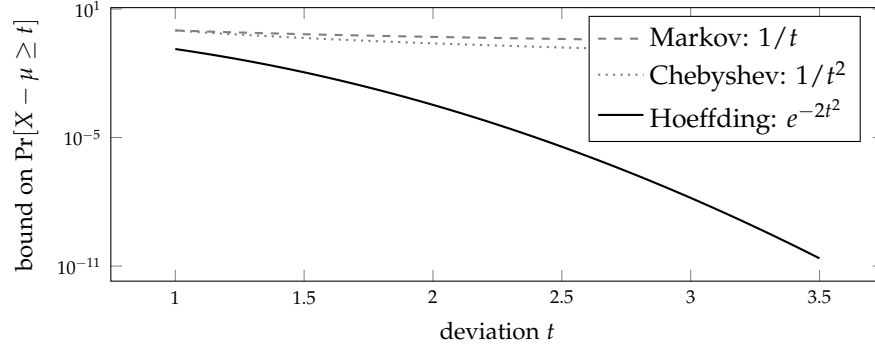


Figure 1.1: Tail decay for a unit-scale random quantity: moment bounds fall polynomially, independence buys the exponential curve — the gap that makes $\text{poly}(n)$ -way union bounds affordable.

Theorem 1.4 — Chernoff, multiplicative form. Let $X_1, \dots, X_m$ be independent random variables with $X_i \in [0, 1]$, $X = \sum_{i=1}^m X_i$, $\mu = \mathbb{E}X$. Then for every $\varepsilon \in (0, 1)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/3}, \quad \Pr[X \leq (1 - \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/2}. \quad \blacksquare$$

Proof idea. Chernoff method (Lemma 1.3) on X , then the elementary bound $(1 + \varepsilon)\ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ (and its lower-tail twin) on the optimized exponent; the full derivation is in the Appendix (Section 1.5). $\blacksquare$

The additive form is interchangeable whenever the summands are bounded but not naturally counts [6, 11, 13]:

Theorem 1.5 — Hoeffding, additive form. Let $X_1, \dots, X_m$ be independent with $X_i \in [a_i, b_i]$ a.s. and $X = \sum_i X_i$. Then for every $t > 0$,

$$\begin{aligned} \Pr[X - \mathbb{E}X \geq t] &\leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right), \\ \Pr[X - \mathbb{E}X \leq -t] &\leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right). \end{aligned} \quad \blacksquare$$

Proof idea. Chernoff method plus *Hoeffding's lemma*, $\mathbb{E}[e^{\lambda(X_i - \mathbb{E}X_i)}] \leq e^{\lambda^2(b_i - a_i)^2/8}$; optimize λ (proof in Section 1.5). $\blacksquare$

1

When independence fails but each random choice shifts the running statistic by a bounded amount, the ladder has two more rungs — Azuma–Hoeffding and McDiarmid — built on martingales; they are stated, with usage guidance, in Section 1.4.

1.0.2 How to choose a rung

The selection procedure is mechanical once two questions are answered. *First, what depends on what?* If the random object is a sum of independent

¹ The entry fee for both forms is *independence plus bounded summands*. The tunable currency is the exponent $-\varepsilon^2\mu$ or $2t^2 / \sum_i (b_i - a_i)^2$ — which must be pushed past $\log(\#\text{events}/\delta)$ for the failure probability to amortize. When the summands are unbounded but light-tailed (Gaussian, Laplace, sub-exponential, sub-gamma), the same Chernoff-method machinery applies with a moment-generating function that exists near 0; corpus papers use sub-gamma tails for approximate counters (Morris-type) and Laplace tails for privacy noise, and we will see the latter in Section 1.4.2.

bounded indicators — samples retained, points in a region, hashed keys — use multiplicative Chernoff when the bound should be relative to the expectation μ (sampling rates, subsampling), and additive Hoeffding when it should be relative to the range parameters $b_i - a_i$ (weighted sums, real-valued measurements). *Second, what moments are available?* If only the mean (or an exponent-optimized mean of a transform) is computable, Markov is the floor — and, applied to $e^{\lambda X}$, it is secretly the ceiling too.

When to Reach for Concentration

Reach whenever a randomized component must be certified — sampling/sketching estimators, random walks, sampled data structures, hashing, privacy noise — and the deliverable is a probability-qualified contract: accuracy $(1 \pm \varepsilon)$ whp, or time/space bounds holding with probability $1 - 1/N$. Then run the recipe:

1. Name the random object X and the bad event $\{|X - \mathbb{E}X| \geq t\}$.
2. Check the dependence structure: independent bounded summands $\rightarrow$ Chernoff/Hoeffding; adaptive exposure or capped per-variable influence $\rightarrow$ the martingale rungs of Section 1.4; only moments available $\rightarrow$ Markov/Chebyshev — constant probability, boostable by repetition with medians (and Markov on $e^{\lambda X}$ is the engine of every exponential-tail argument anyway).
3. Tune parameters so each bad event fails with probability at most $\delta/(\text{events})$ — i.e. push the exponent past $\log(\text{events}/\delta)$; only exponential tails amortize over $\text{poly}(n)$ copies, polynomial ones do not.
4. Union-bound over all copies of the component.

1.1 Trimmed statistics in a turnstile stream

The exemplar is a PODS'26 paper on robust streaming statistics [8]^{PODS}. Frequency moments, $F_p = \sum_i |x_i|^p$, are the workhorses of database analytics: query optimization, data mining, and network monitoring all rely on them, and the classical streaming results estimate them over the *whole* frequency vector. Many analytics tasks, however, want order-restricted quantities instead: statistics built from only the heaviest (or the central) part of the data, with the extremes treated as outliers — the streaming analogue of the trimmed mean from robust statistics. The paper studies such *trimmed statistics* — notably the F_p moment of the top- k coordinates, a Ky-Fan-type norm behind impact measures like the h -index — in the turnstile streaming model, and shows that for every

$p \in [0, 2]$ they can be estimated in a single pass using $\text{poly}(\log n / \varepsilon)$ bits of space.

Problem. Work in the *strict turnstile* streaming model: an n -dimensional frequency vector $\mathbf{x} \in \mathbb{Z}^n$ starts at zero and receives a long stream of updates of the form “add Δ to coordinate i ” (with Δ of either sign); the algorithm sees updates one at a time and must answer queries about the final $\mathbf{x}$ using space far less than n . Let $a_1, \dots, a_k$ be the k coordinates of $\mathbf{x}$ of largest absolute value. The *trimmed statistic* of order (p, k) is $\sum_{i=1}^k |a_i|^p$ for $0 \leq p \leq 2$: the F_p mass contributed by the top k coordinates — the part of the data that survives trimming away the tail. Such statistics are the streaming analogue of trimmed estimators in robust statistics.

Guarantee.

Theorem 1.6 — Top- k F_p estimation, Theorem 4 of the paper. Let $\mathbf{x} \in \mathbb{Z}^n$, $p \in [0, 2]$, $k \geq 0$, $\varepsilon \in (0, 1)$. Suppose $\|\mathbf{x}_{-k}\|_2^2 \geq (\varepsilon / \log n)^c \cdot k$ for a constant c , where $\mathbf{x}_{-k}$ excludes the top- k coordinates. Then there exists a *linear sketch* using $\text{poly}(\log n / \varepsilon)$ bits of space that estimates $\sum_{i=1}^k |a_i|^p$ to a $(1 \pm \varepsilon)$ factor with high constant probability. ■

2

How concentration is applied. The algorithm partitions coordinates by magnitude into *geometric level sets* whose boundaries are shifted by a random offset ζ uniform in $[1/2, 1]$, then *dyadically subsamples* each level: each coordinate of level S_j is kept independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$ where $s_j = |S_j|$ and C is a constant. Survivors are found with a CountSketch³ fed with $\Theta(\log n)$ buckets per subsampling rate. The concentration step is the following lemma; its random objects are the independent retention indicators X_i of the coordinates of S_j , each Bernoulli(r), so the survivor count has mean $s_j r$.

Lemma 1.7 — Level-size estimation, Lemma 4.4 of the paper, notation cleaned. Let S_j be a level set with $s_j = |S_j|$, and subsample each coordinate independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$. If z coordinates of S_j survive and $\tilde{s}_j = z/r$, then with probability $1 - 1/\text{poly}(n)$,

$$(1 - \varepsilon) s_j \leq \tilde{s}_j \leq (1 + \varepsilon) s_j. \quad \blacksquare$$

Proof sketch. Write $u_1, \dots, u_{s_j}$ for the coordinates of S_j , and let X_i be the indicator that u_i is sampled, so $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff gives $\Pr[|\sum_i X_i - s_j r| \geq \varepsilon s_j r] \leq 2 \exp(-\varepsilon^2 s_j r / 3) \leq 2 \exp(-C \log n / 3) \leq 1 / \text{poly}(n)$; divide by r . (Full proof in Section 1.5.) ■

² A *linear sketch* stores $A\mathbf{x}$ for a random matrix A ; it is the standard route to mergeable, one-pass, deletion-tolerant stream summaries. The condition on $\|\mathbf{x}_{-k}\|_2^2$ holds automatically once $k \geq \text{polylog } n$, so the theorem covers the practically interesting regime.

³ CountSketch (Charikar–Chen–Farach-Colton) recovers, from hashed counters, all coordinates whose value exceeds a θ -fraction of the residual ℓ_2 norm (a *heavy hitter* structure); the paper needs only this black box.

The δ tuning is explicit: r is chosen exactly so that the Chernoff exponent $\epsilon^2 s_j r$ equals $C \log n$, i.e. per-level failure $1/\text{poly}(n)$; there are only $O(\log_{1+\epsilon} m)$ levels (magnitude scales), so a union bound over levels preserves the polynomial failure probability. A companion lemma (Lemma 4.5 of the paper) uses the same Chernoff bound on the binomial survivor count to show that fewer than $O(\log n / \epsilon^2)$ coordinates survive per level whp — which is what makes the heavy-hitter structure, and hence the space bound, $\text{poly}(\log n / \epsilon)$.

What it buys. The system gets a one-pass, sublinear-space summary of trimmed statistics — robust aggregate mass that ignores the tail — with a certificate strong enough to union-bound into larger algorithms; the randomness of the offset ζ (rather than an adversarial grid) is what makes the level-set decomposition itself immune to worst-case input placement.

1.2 Maximum range sum by randomized grid shifting

The exemplar is a SIGMOD’26 paper on spatial query processing [5]. The *maximum range sum* (MaxRS) problem sits behind every “find the best location” query: given weighted points — customers, incidents, sensors — and a fixed-size geometric region Q , place Q anywhere in space so that it captures as much total weight as possible; that placement is where to site a store, a base station, or an emergency facility. The problem is well studied by both the database and the computational-geometry communities, and the paper revisits it in three practically motivated variations: *dynamic* MaxRS, where points are inserted and deleted and a good placement must be maintained under updates; *batched* MaxRS, where many query regions must be answered at once; and *colored* variants that group the points into classes. Its headline results are a randomized $(\frac{1}{2} - \epsilon)$ -approximation for dynamic MaxRS with polylogarithmic update time — the first algorithm of this kind — together with conditional lower bounds showing that the batched problem cannot be solved substantially faster than the trivial approach.

Problem. In the *maximum range sum* (MaxRS) problem, the input is a set P of n points in $\mathbb{R}^d$, each carrying a weight, and a query shape Q — here a d -ball of given radius. The goal is to place Q (translate it anywhere) so as to maximize the total weight of the points of P covered.

Exact MaxRS is expensive: for Q a 2-ball the colored version carries a conditional $\Omega(n^2)$ lower bound, and assuming that $(\min, +)$ -convolution on length- n sequences requires $\Omega(n^2)$ time, the batched variant on n points and m query radii requires $\Omega(mn)$ time — almost matching the trivial $O(mn \log n)$ upper bound.

Guarantee.

Theorem 1.8 — Static MaxRS, one representative case of the paper. For static MaxRS with Q a d -ball and any $0 < \varepsilon < 1/2$, there is a randomized $(\frac{1}{2} - \varepsilon)$ -approximation algorithm running in time $O(\varepsilon^{-2d-2} n \log n)$; the approximation factor holds with high probability, i.e. $1 - n^{-\Omega(1)}$. The dynamic version achieves the same factor with amortized update time $O(\varepsilon^{-2d-2} \log n)$. ■

How concentration is applied. The algorithm randomly shifts a grid over the point set and, per cell, takes a random sample of the cell's points; the candidate balls it examines are drawn from a finite set of $k \leq n$ possibilities determined by the shift. Two facts combine. First, a *geometric covering lemma*: if C is a d -ball of radius ρ and B a unit d -ball whose center lies at distance ρ^2 from the center of C , then the $(d-1)$ -dimensional measure of the cap $B \cap \partial C$ is at least a $(\frac{1}{2} - \Theta(\varepsilon))$ fraction of the measure of ∂C — i.e., a random point of C 's boundary lands in any fixed candidate ball with probability at least $\frac{1}{2} - \Theta(\varepsilon)$. Second, the concentration step:

Lemma 1.9 — Sampling concentration, Lemma of the paper, notation cleaned. Let $S = \{p_1, \dots, p_t\}$ be such that each point of S lies inside the candidate ball B independently with probability at least $\frac{1}{2} - \Theta(\varepsilon)$, and let $Y_B = \sum_{i=1}^t \mathbf{1}\{p_i \in B\}$. Then with high probability, simultaneously for all $k \leq n$ candidate balls,

$$Y_B \geq (1 - \varepsilon) \left(\frac{1}{2} - \Theta(\varepsilon) \right) t. \quad \blacksquare$$

Proof sketch. The $X_i = \mathbf{1}\{p_i \in B\}$ are independent indicators with $\mathbb{E}[Y_B] \geq (\frac{1}{2} - \Theta(\varepsilon))t$; multiplicative Chernoff gives $\Pr[Y_B < (1 - \varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 t/2}$, and a union bound over the $k \leq n$ balls keeps this whp provided $t = \Theta(\varepsilon^{-2} \log n)$. (Full proof in Section 1.5.) ■

The random objects are the per-point sampling/shift indicators; the structure is full independence across points (the random shift acts independently per cell); the rung is multiplicative Chernoff; and the δ tuning sets $t = \Theta(\varepsilon^{-2} \log n)$ so each of the n candidate balls fails with probability $n^{-\Omega(1)}$ — the same “push the exponent past $\log(\text{events})$ ” accounting as in Section 1.1, with the exponent coming from sample size rather than subsample rate.

What it buys. The spatial engine answers best-placement queries in near-linear time with a certified $(1/2 - \varepsilon)$ fraction of the optimum whp — and supports updates in polylog amortized time — where

⁴ *Conditional lower bound:* a reduction from $(\min, +)$ -convolution — widely conjectured to need quadratic time — transfers that hardness to MaxRS. This is the paper's evidence that *approximation* is necessary, and approximation is exactly where concentration enters.

exact computation faces (conditional) quadratic barriers. The same Chernoff-plus-union-bound skeleton, re-instantiated with trapezoidal maps, gives the paper’s $(1 - \epsilon)$ -approximation for the colored variant.

1.3 User-level private query processing with sampled tuples

The exemplar is a SIGMOD’26 paper on private query processing [12]. Answering select–join–aggregate (SJA) queries under differential privacy is a basic building block of privacy-preserving analytics: the system must return an accurate aggregate while guaranteeing that no adversary can confidently infer whether any one *user’s* data was present or absent. The state of the art achieves strong, instance-optimal accuracy at this *user level*, but pays for it by solving expensive optimization programs over the whole database — prohibitive at scale. Sampling is the classic escape hatch, buying a tunable trade-off between accuracy and computational cost, yet combining it with private processing is subtle: it is unclear *what* to sample for the best accuracy under a given compute budget, and the existing mathematical tool chains were not designed with sampling in mind. The paper builds a scalable user-level-DP SJA engine that privately truncates each heavy user’s contribution and Poisson-samples tuples to accelerate the truncation search, with an error analysis that cleanly separates sampling error from privacy noise.

Problem. Consider a select–project–join–aggregate workload over a relational instance whose rows belong to *users*, each user contributing many tuples. The privacy contract is differential privacy at the granularity of users:

Definition 1.10 (ϵ -differential privacy, user level). A randomized mechanism M satisfies ϵ -differential privacy if for all *neighboring* inputs D, D' and all measurable sets S ,

$$\Pr[M(D) \in S] \leq e^\epsilon \Pr[M(D') \in S].$$

For *user-level* DP, “neighboring” means *all* tuples of one user added or removed — much stricter than tuple-level DP.

Unbounded per-user contributions then blow up sensitivity, so the system [12] truncates each user’s contribution to a threshold τ chosen privately by a search (sparse vector technique), and accelerates the search by Poisson-sampling tuples at rate q before privatizing.

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⁵ Poisson sampling also *amplifies* privacy: a mechanism that is ϵ -DP on the sample is (ϵ') -DP on the full data with $\epsilon' = \ln(1 + q(e^\epsilon - 1)) < \epsilon$; the paper’s Sample-and-Explore comparison baselines use this. The proof is a binomial tail computation — again concentration.

Guarantee.

Theorem 1.11 — Error of the sampled truncation, Algorithm 2 of the paper, compressed. The mechanism satisfies $(\varepsilon, 0)$ -DP. Let n be the number of records, m the number of users, and q the sample rate. Write $\tau^*(T) = \max_u \sum_{t \in T} \mathbf{1}[u \prec t]$ for the true maximum per-user contribution, and $\varepsilon_i^* \geq \varepsilon/L$ for the noise-calibration budget over the L search iterations. With probability $1 - \beta$, the released answer $\hat{f}(T)$ obeys

$$|\hat{f}(T) - f(T)| = \tilde{O}\left(\sqrt{\frac{n}{q}} + \frac{1}{q} + \frac{1}{\varepsilon_i^*} \left(\tau^*(T) + \sqrt{\frac{\tau^*(T)}{q}}\right)\right),$$

where $\tilde{O}(\cdot)$ hides constants and polylogarithmic factors in m and $1/\beta$. ■

How concentration is applied. The random objects are the independent sampling indicators $\mathbf{1}\{t \text{ sampled}\} \sim \text{Bernoulli}(q)$ over the n tuples. The sampled statistic $f(S) = \sum_{t \in T} w_t \mathbf{1}[t \text{ sampled}]$ satisfies $\mathbb{E}[f(S)] = q f(T)$ and $\text{Var}[f(S)] = q(1-q) \sum_t w_t^2 \leq qn$: a sum of independent bounded variables with an explicit variance proxy — the territory of the Bernstein/Chernoff rung, whose bound gives $|f(S) - qf(T)| \lesssim \sqrt{qn \log(3L/\beta)}$ with probability $1 - \beta/(3L)$; dividing by q yields the $\sqrt{n/q}$ term. The δ -budgeting is again structural: the truncation-threshold search runs L iterations, each is allocated failure probability $\beta/(3L)$, and a union bound over iterations upgrades the per-iteration bound to the global $1 - \beta$ contract — with the same per-event log factor $\log(3L/\beta)$ appearing inside the noise scale. The $\tau^*(T)/\varepsilon_i^*$ terms come from the Laplace-style calibration of the final release against the (privately learned) per-user cap.

What it buys. The query engine can offer user-level — not merely tuple-level — privacy at a modest accuracy price: the sampling error $\sqrt{n/q}$ decouples from the privacy noise, so tuning q trades compute for accuracy without touching the privacy guarantee, and the explicit $\tau^*(T)$ dependence quantifies exactly how much heavy user-contributors cost the answer.

1.4 Advanced usage and further reading

Everything so far needs independence. Two extensions belong in every practitioner’s backpack: the martingale rungs that replace independence with bounded per-step influence, and the light-tailed regime where summands are unbounded but have a moment generating function. This section states the rungs, sketches two harder exemplars, and points to the literature.

⁶ Recall: a martingale difference sequence (Y_i) with respect to a filtration $(\mathcal{F}_i)$ has $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0$; the partial sums $M_j = \sum_{i \leq j} Y_i$ then satisfy $\mathbb{E}[M_j \mid \mathcal{F}_{j-1}] = M_{j-1}$.

1.4.1 Martingale rungs: Azuma–Hoeffding and McDiarmid

Expose the randomness sequentially and book the shifts as a *martingale*⁶:

Theorem 1.12 — Azuma–Hoeffding. Let $(Y_i)_{i=1}^m$ be a martingale difference sequence with $|Y_i| \leq c_i$ a.s. Then for every $t > 0$,

$$\Pr \left[\sum_{i=1}^m Y_i \geq t \right] \leq \exp \left(- \frac{t^2}{2 \sum_{i=1}^m c_i^2} \right). \quad \blacksquare$$

Proof idea. Chernoff method on $\sum_i Y_i$, Hoeffding’s lemma applied to each Y_i *conditionally on the past* (boundedness is all that is needed; independence is replaced by iterated conditioning), then optimize λ (Section 1.5). $\blacksquare$

The function-of-independent-variables form is the rung most used for adaptive algorithms and data structures, where the analyzed quantity is not a sum but a function whose per-variable influence is capped [2, 4, 10]:

Theorem 1.13 — McDiarmid, bounded differences. Let $X_1, \dots, X_m$ be independent random variables and $f: \prod_i \mathcal{X}_i \rightarrow \mathbb{R}$ a measurable function satisfying the bounded differences condition: for each i there is $c_i \geq 0$ with

$$|f(x_1, \dots, x_i, \dots, x_m) - f(x_1, \dots, x'_i, \dots, x_m)| \leq c_i$$

for all inputs and all x'_i . Then for every $t > 0$,

$$\Pr [f(X_1, \dots, X_m) - \mathbb{E}f \geq t] \leq \exp \left(- \frac{2t^2}{\sum_{i=1}^m c_i^2} \right). \quad \blacksquare$$

Proof idea. Apply Azuma–Hoeffding (Theorem 1.12) to the *Doob martingale* $Y_i = \mathbb{E}[f \mid X_1, \dots, X_i] - \mathbb{E}[f \mid X_1, \dots, X_{i-1}]$, whose increments are bounded by c_i exactly because changing X_i moves f by at most c_i (Section 1.5). $\blacksquare$

When these rungs earn their keep. If the algorithm is *adaptive* — later random choices conditioned on earlier estimates, sampling without replacement, stopping times — the summands are dependent but each step’s effect is bounded, and Azuma–Hoeffding applies. If the quantity is a *function* of many independent inputs (a data structure’s quality, a clustering cost, a query plan’s runtime) with per-input influence at most c_i , McDiarmid converts that directly. Hashing with limited independence is the classic gap: the indicators are not fully independent, but exposure by hash key gives bounded increments.

1.4.2 Two advanced exemplars, sketched

Both exemplars of this section are privacy papers that push past the sum-of-independent-bounded-summands regime; each is sketched in

one paragraph, and the round-by-round argument of the first is transcribed in the Appendix (Section 1.5).

Private graph pattern counting under LDP [7]. Counting subgraph patterns — triangles, paths, stars — is a cornerstone of network analysis (community detection, fraud detection), and releasing such counts leaks information about individual edges. Under *local* differential privacy there is no trusted curator: each node of the graph randomizes its own messages before anything is aggregated, and existing LDP mechanisms were ad hoc, handling only specific patterns such as triangles and stars. The paper gives the first general mechanism for counting arbitrary *acyclic* patterns under LDP. It runs k rounds of message passing, adding fresh Laplace noise — *sub-exponential* tails, so the Chernoff method still applies — at every node in every round. The δ -accounting is the recipe of Section 1.0.2 in its purest form: failure budget β'/N per node per round, union bound over the kN node-rounds, then an *induction on rounds* that grows the noise scale geometrically but controllably. The resulting estimate is unbiased with error $\tilde{O}(\sqrt{N d(G)^{k-1}})$ — the square root of the maximal possible count. For simple paths, the tail is only polynomial in $1/\beta$ (a variance/Markov argument), illustrating the price of dropping exponential tails.

Product noise as a concentration bound [9]. Adding calibrated noise to a released statistic is the most fundamental route to differential privacy: the system outputs $f(x) + \mathbf{n}$, and the guarantee rests on the tail of the *privacy-loss random variable* $L_M = \ln \frac{dP}{dQ}$ — (ϵ, δ) -DP holds exactly when $\Pr[L_M \geq \epsilon] \leq \delta$, a tail bound in disguise. For Gaussian noise L_M is itself a univariate Gaussian, and this is where standard analyses stop. The paper instead exploits the spherical symmetry of typical noise distributions to generate $\mathbf{n}$ from a *product* of two independent components — a radius controlling the noise magnitude and a direction controlling its angle relative to the sensitivity vector — which re-expresses L_M as a product measure and lets the tail be optimized directly. The paper never writes the word “Chernoff,” but its argument — factorize L_M into independent terms, bound $\mathbb{E}[U^q]$ for a transform U , and *minimize the resulting δ over the exponent q* — is exactly the Chernoff method (Lemma 1.3) on a likelihood ratio. The payoff: expected noise smaller by a $\Theta(\log(1/\delta))$ factor than the Gaussian mechanism at equal privacy.

1.4.3 Further reading

- Boucheron, Lugosi, and Massart [2] — the comprehensive non-asymptotic treatment; the martingale and supremum versions live here.

- Dubhashi and Panconesi [4] — the randomized-algorithms viewpoint, with the exposure/Doob-martingale technique foregrounded.
- Mitzenmacher and Upfal [11] — textbook treatment with CS applications and all the Chernoff variants used in systems papers.
- The originals: Chernoff (1952) [3], Hoeffding (1963) [6], Azuma (1967) [1], and McDiarmid’s bounded-differences survey [10].
- For a two-page refresher with proofs of the additive forms, the Stanford CS229 notes [13].

Chapter Recap

- **What the tool is.** Concentration inequalities convert expectations into with-high-probability contracts: $\Pr[|X - \mathbb{E}X| \geq t]$ bounded by a tail function, tuned so that per-event failure $\delta/(n)$ survives a union bound over all copies of the randomized component.
- **Which rung when.** Independent bounded indicators/sums: multiplicative Chernoff (relative error) or additive Hoeffding (range-based). Adaptive or dependent exposure with bounded per-step effect: Azuma–Hoeffding. Function of many independent inputs with capped per-input influence: McDiarmid. Only moments available: Markov and Chebyshev — constant-probability results, boostable by medians. Light-tailed unbounded noise (Laplace, Gaussian, sub-gamma): the Chernoff method still applies via the moment generating function.
- **Why exponential tails.** Only $e^{-\Omega(\text{exponent})}$ tails make $\text{poly}(n)$ -way union bounds affordable; polynomial tails ($1/t$, $1/t^2$) cannot amortize — visible in the corpus as the gap between $\tilde{O}(\cdot)$ guarantees (Laplace/Chernoff paths) and $1/\beta$ -type errors (variance-only paths).
- **What the corpus showed.** Trimmed-statistic sketching (Section 1.1): Chernoff on subsampling survivors, union over $O(\log_{1+\epsilon} m)$ levels. Spatial MaxRS (Section 1.2): Chernoff on sampled point-in-ball indicators, union over $\leq n$ candidate balls. User-level private queries (Section 1.3): Bernstein/Chernoff on Bernoulli-sampled sums with $\beta/(3L)$ budgets over search iterations.
- **Advanced usage.** The martingale rungs (Section 1.4.1) replace independence with bounded per-step influence; the privacy exemplars of Section 1.4.2 run the Chernoff method on Laplace

noise and on a likelihood ratio. Long proofs of everything deferred live in the Appendix (Section 1.5).

1.5 Appendix

Proofs omitted from the main text are collected here.

Proof of Theorem 1.4 (Chernoff, multiplicative form)

Statement (recalled). Let $X_1, \dots, X_m$ be independent with $X_i \in [0, 1]$, $X = \sum_i X_i$, $\mu = \mathbb{E}X$. Then $\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/3}$ and $\Pr[X \leq (1 - \varepsilon)\mu] \leq e^{-\varepsilon^2\mu/2}$. ■

Proof. By the Chernoff method (Lemma 1.3), for $\lambda > 0$, $\Pr[X \geq (1 + \varepsilon)\mu] \leq e^{-\lambda(1+\varepsilon)\mu} \prod_i \mathbb{E}[e^{\lambda X_i}]$. For $X_i \in [0, 1]$ independence gives $\mathbb{E}[e^{\lambda X_i}] = 1 + p_i(e^\lambda - 1) \leq \exp(p_i(e^\lambda - 1))$, so the product is at most $e^{\mu(e^\lambda - 1)}$ and, choosing $\lambda = \ln(1 + \varepsilon)$,

$$\Pr[X \geq (1 + \varepsilon)\mu] \leq \left(\frac{e^\varepsilon}{(1 + \varepsilon)^{1+\varepsilon}} \right)^\mu \leq e^{-\varepsilon^2\mu/3},$$

using $(1 + \varepsilon)\ln(1 + \varepsilon) - \varepsilon \geq \varepsilon^2/3$ for $\varepsilon \in (0, 1)$. The lower tail is symmetric, with exponent $\varepsilon^2/2$. ■

Proof of Theorem 1.5 (Hoeffding, additive form)

Statement (recalled). Let $X_1, \dots, X_m$ be independent with $X_i \in [a_i, b_i]$ a.s. and $X = \sum_i X_i$. Then for every $t > 0$, $\Pr[X - \mathbb{E}X \geq t] \leq \exp(-2t^2 / \sum_i (b_i - a_i)^2)$, and the same for the lower tail. ■

Proof. Hoeffding's lemma: if $Y \in [a, b]$ a.s. and $\mathbb{E}Y = 0$, then $\mathbb{E}[e^{\lambda Y}] \leq e^{\lambda^2(b-a)^2/8}$. (Proof: $e^{\lambda y}$ is convex, so $e^{\lambda Y} \leq \frac{b-Y}{b-a}e^{\lambda a} + \frac{Y-a}{b-a}e^{\lambda b}$ pointwise; take expectations and minimize over a shifted λ .) Applying the lemma to each centered summand and multiplying by independence, $\mathbb{E}[e^{\lambda(X - \mathbb{E}X)}] \leq e^{\lambda^2 \sum_i (b_i - a_i)^2/8}$; the Chernoff method then gives $\Pr[X - \mathbb{E}X \geq t] \leq e^{-\lambda t + \lambda^2 \sum_i (b_i - a_i)^2/8}$, and optimizing at $\lambda = 4t / \sum_i (b_i - a_i)^2$ yields the stated $e^{-2t^2 / \sum_i (b_i - a_i)^2}$. ■

Proof of Theorems 1.12 and 1.13 (martingale rungs)

Statement (recalled). (i) If $(Y_i)_{i=1}^m$ is a martingale difference sequence with $|Y_i| \leq c_i$ a.s., then $\Pr[\sum_i Y_i \geq t] \leq \exp(-t^2/2 \sum_i c_i^2)$. (ii) If f has bounded differences (c_i) over independent $X_1, \dots, X_m$, then $\Pr[f - \mathbb{E}f \geq t] \leq \exp(-2t^2 / \sum_i c_i^2)$. ■

Proof. Let $M_j = \sum_{i \leq j} Y_i$ with $\mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] = 0$ and $|Y_i| \leq c_i$. Iterated conditioning gives $\mathbb{E}[e^{\lambda M_m}] = \mathbb{E}[e^{\lambda M_{m-1}} \mathbb{E}[e^{\lambda Y_m} \mid \mathcal{F}_{m-1}]]$, and *conditionally on $\mathcal{F}_{m-1}$* the variable Y_m is centered and bounded in an interval of length $2c_m$, so Hoeffding's lemma applies with $(b-a) = 2c_m$. Unrolling, $\mathbb{E}[e^{\lambda M_m}] \leq e^{\lambda^2 \sum c_i^2 / 2}$, and optimizing λ gives the stated tail. For McDiarmid, set $Y_i = \mathbb{E}[f \mid X_1, \dots, X_i] - \mathbb{E}[f \mid X_1, \dots, X_{i-1}]$: this is a martingale difference sequence with $|Y_i| \leq c_i$ precisely because resampling only X_i moves f by at most c_i (take conditional expectations over the two values of X_i), and $\sum_i Y_i = f - \mathbb{E}f$. Azuma–Hoeffding applied to (Y_i) is McDiarmid's inequality. ■

Proof of Lemma 1.7 (level-size estimation, Section 1.1)

Statement (recalled). Subsample each coordinate of a level set S_j , $|S_j| = s_j$, independently with probability $r = \min\{1, C \log n / (\varepsilon^2 s_j)\}$. If z coordinates survive, then $\tilde{s}_j = z/r$ satisfies $(1 - \varepsilon)s_j \leq \tilde{s}_j \leq (1 + \varepsilon)s_j$ with probability $1 - 1/\text{poly}(n)$. ■

Proof. Denote the coordinates of S_j by u_i , $i \in [s_j]$, and let X_i be the indicator that u_i is sampled into substream P , so the X_i are independent with $\mathbb{E}[X_i] = r$ and $\mathbb{E} \sum_i X_i = s_j r$. Multiplicative Chernoff (Theorem 1.4) gives

$$\begin{aligned} \Pr\left[\left|\sum_i X_i - s_j r\right| \geq \varepsilon s_j r\right] &\leq 2 \exp(-\varepsilon^2 s_j r / 3) \leq 2 \exp(-C \log n / 3) \\ &\leq 1 / \text{poly}(n), \end{aligned}$$

where the second inequality uses $s_j r = C \log n / \varepsilon^2$ (when the minimum in the definition of r is attained at this branch; otherwise $r = 1$ and there is nothing to estimate). Dividing by r converts the event into $(1 - \varepsilon)s_j \leq \tilde{s}_j \leq (1 + \varepsilon)s_j$. ■

Proof of Lemma 1.9 (sampling concentration, Section 1.2)

Statement (recalled). If each point of $S = \{p_1, \dots, p_t\}$ lies in the candidate ball B independently with probability at least $\frac{1}{2} - \Theta(\varepsilon)$, then simultaneously for all $k \leq n$ candidate balls, $Y_B = \sum_i \mathbf{1}\{p_i \in B\} \geq (1 - \varepsilon)(\frac{1}{2} - \Theta(\varepsilon))t$ whp. ■

Proof. Fix a candidate ball B and let $X_i = \mathbf{1}\{p_i \in B\}$; the X_i are independent, and by the covering property $\mathbb{E}[Y_B] = \sum_i \mathbb{E}X_i \geq (\frac{1}{2} - \Theta(\varepsilon))t$. Multiplicative Chernoff gives

$$\Pr[Y_B < (1 - \varepsilon)\mathbb{E}Y_B] \leq e^{-\varepsilon^2 \mathbb{E}Y_B / 2} \leq e^{-\Theta(\varepsilon^2 t)}.$$

With $t = \Theta(\varepsilon^{-2} \log n)$ this is $n^{-\Omega(1)}$ per ball, and a union bound over the $k \leq n$ candidate balls gives the simultaneous statement $Y_B \geq (1 - \varepsilon)(\frac{1}{2} - \Theta(\varepsilon))t$ for all of them, with high probability. ■

Round-by-round noise control (Section 1.4.2)

Statement (recalled). The k -round walk mechanism of [7] is unbiased for the walk count W_k and, with probability $1 - \beta$, errs by at most $k\gamma\sqrt{N}(d(G) + \gamma)^{k-1}$, where $\gamma = 2k\sqrt{8\log(2kN/\beta)}/\varepsilon$. ■

Proof. Write $X_i^{(\ell)}$ for node i 's round- ℓ value. The mechanism gives, for each node i and round $\ell \in \{1, \dots, k\}$,

$$X_i^{(\ell)} = \sum_{j \in N(i)} X_j^{(\ell-1)} + \text{Lap}(2k\|X^{(\ell-1)}\|_\infty/\varepsilon),$$

so $\sum_{j \in N(i)} X_j^{(\ell-1)} \leq d(G)\|X^{(\ell-1)}\|_\infty$ always. Allocate failure β'/N per node per round: the Laplace tail $\Pr[\text{Lap}(b) > u] \leq \frac{1}{2}e^{-u/b}$ gives, with probability $1 - \beta'/N$ per node-round, $2k\text{Lap}(\|X^{(\ell-1)}\|_\infty/\varepsilon) \leq \gamma\|X^{(\ell-1)}\|_\infty$ for the stated γ (with $\beta' \approx \beta/k$). Union-bounding over the $\leq kN$ node-rounds, all these events hold simultaneously with probability $\geq 1 - (k-1)\beta'$, and on that event an induction gives

$$\|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)\|X^{(\ell-1)}\|_\infty \implies \|X^{(\ell)}\|_\infty \leq (d(G) + \gamma)^\ell$$

for all $\ell \leq k-1$. The released count aggregates the N per-node values, whose centered Laplace contributions are independent sub-exponential variables each of scale at most $O(\gamma(d(G) + \gamma)^{k-1})$; their sum concentrates around zero at scale $\sqrt{N} \cdot \tilde{O}(\gamma(d(G) + \gamma)^{k-1})$ by the same Chernoff-method argument applied to light-tailed variables, which yields the stated $k\gamma\sqrt{N}(d(G) + \gamma)^{k-1}$ error radius with probability $1 - \beta$. ■

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